THE SIMPLICIAL RESOLUTION OF O ON THE TOTAL SITE (RINGED CASE)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the homotopy-invariance input (Delta[m] simeq Delta[0], cited from the Simplicial chapter at source lines 2124-2132) and the functoriality of taking free abelian sheaves are the only ingredients invoked; the adjoint description of the differentials uses the g_{n!} description established on cards L46149 and L46168; no verification is omitted.

Editorial note. The label lemma-simplicial-resolution-ringed names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The simplicial-site situation with a sheaf of rings. Same chapter, Situation \label{situation-simplicial-site}. Statement lines 2093-2095: O is a sheaf of rings on C_total; g_{n!} is as in lemma-restriction-module-to-components-site (card L46168).
- The description of the functors g_{n!}. Same chapter, \label{lemma-restriction-to-components-site} (card L46149) and \label{lemma-restriction-module-to-components-site} (card L46168). Proof lines 2103-2105: the proof uses the description of g_{n!} from the restriction lemmas.
- The free module on a simplicial set and homotopy invariance. Simplicial, Example \label{simplicial-example-simplex-simplicial-set}, Example \label{simplicial-example-simplex-contractible}, Remark \label{simplicial-remark-homotopy-better} and \label{simplicial-lemma-homotopy-equivalence-s-N}. Proof lines 2114-2132: g_m^{-1} of the complex is the free O_m-module on the simplicial set Delta[m]; Delta[m] is homotopy equivalent to Delta[0] and taking free abelian sheaves is functorial.

Source reference. Original reference 0D72: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1459–1459. Prerequisite pointer invoked at proof line(s) 2100; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 2105; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 08Q3: sources/stacks-chow/tags.txt, source lines 11046–11046. Prerequisite pointer invoked at proof line(s) 2124; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0176: sources/stacks-chow/tags.txt, source lines 1569–1569. Prerequisite pointer invoked at proof line(s) 2122; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 019T: sources/stacks-chow/tags.txt, source lines 1664–1664. Prerequisite pointer invoked at proof line(s) 2129; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 019M: sources/stacks-chow/tags.txt, source lines 1657–1657. Prerequisite pointer invoked at proof line(s) 2128; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 2093; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}_{total}$. There is a complex*

$$\ldots \to g_{2!}\mathcal{O}_2 \to g_{1!}\mathcal{O}_1 \to g_{0!}\mathcal{O}_0$$

of $\mathcal{O}$-modules which forms a resolution of $\mathcal{O}$. Here $g_{n!}$ is as in Lemma 0D72.

Proof. We will use the description of $g_{n!}$ given in Lemma 09WG. As maps of the complex we take $\sum(-1)^i d_i^n$ where $d_i^n : g_{n!}\mathcal{O}_n \to g_{n-1!}\mathcal{O}_{n-1}$ is the adjoint to the map $\mathcal{O}_n \to \bigoplus_{[n-1]\to[n]} \mathcal{O}_n = g_n^* g_{n-1!}\mathcal{O}_{n-1}$ corresponding to the factor labeled with $\delta_i^n : [n-1] \to [n]$. Then g_m^{-1} applied to the complex gives the complex

$$\ldots \to \bigoplus\nolimits_{\alpha\in \mathrm{Mor}_\Delta([2],[m])} \mathcal{O}_m \to \bigoplus\nolimits_{\alpha\in \mathrm{Mor}_\Delta([1],[m])} \mathcal{O}_m \to \bigoplus\nolimits_{\alpha\in \mathrm{Mor}_\Delta([0],[m])} \mathcal{O}_m$$

on $\mathcal{C}_m$. In other words, this is the complex associated to the free $\mathcal{O}_m$-module on the simplicial set $\Delta[m]$, see Simplicial, Example 0176. Since $\Delta[m]$ is homotopy equivalent to $\Delta[0]$, see Simplicial, Example 08Q3, and since "taking free abelian sheaf on" is a functor, we see that the complex above is homotopy equivalent to the free abelian sheaf on $\Delta[0]$ (Simplicial, Remark 019M and Lemma 019T). This complex is acyclic in positive degrees and equal to $\mathcal{O}_m$ in degree 0. □